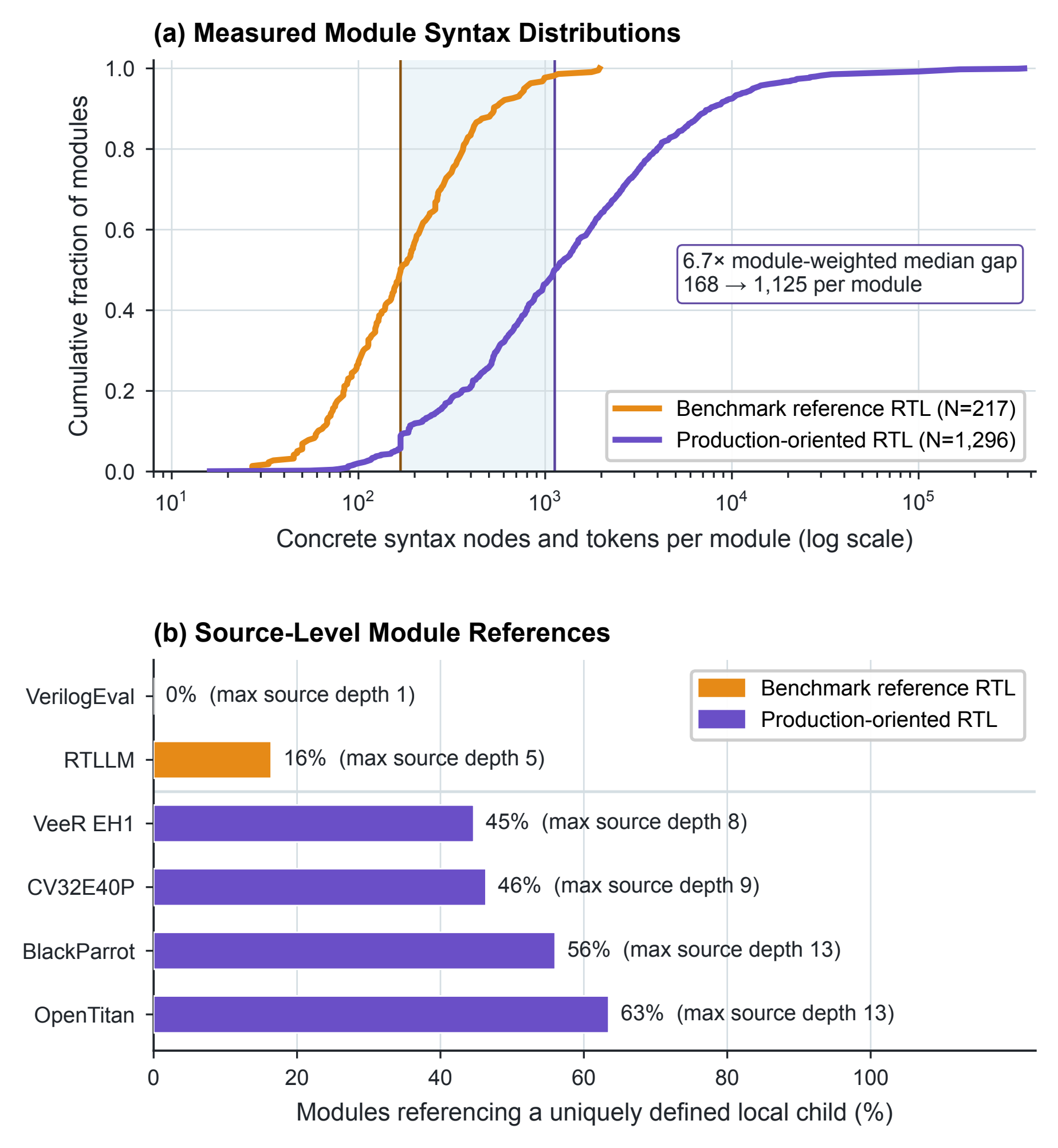
(a) Measured Module Syntax Distributions
Cumulative fraction of modules
1.0
0.8
0.6
0.4
0.2
0.0
6.7× module-weighted median gap
168 → 1,125 per module
Benchmark reference RTL (N=217)
Production-oriented RTL (N=1,296)
10^1
10^2
10^3
10^4
10^5
Concrete syntax nodes and tokens per module (log scale)
(b) Source-Level Module References
Benchmark reference RTL
Production-oriented RTL
VerilogEval
0% (max source depth 1)
RTLLM
16% (max source depth 5)
VeeR EH1
45% (max source depth 8)
CV32E40P
46% (max source depth 9)
BlackParrot
56% (max source depth 13)
OpenTitan
63% (max source depth 13)
0
20
40
60
80
100
Modules referencing a uniquely defined local child (%)